

Bayes' Rule

From conditional probability to Bayes' Rule

Let's start from the **definition of conditional probability**. For any two events A and B with $P(B) > 0$:

$$P(A, B) = P(A | B) \cdot P(B) = P(B | A) \cdot P(A)$$

Both middle and right expressions equal the joint probability $P(A, B)$. Set them equal and divide both sides by $P(B)$:

$$P(A | B) = P(B | A) \cdot P(A) / P(B)$$

That is the entire derivation of the Bayes' rule — one line of algebra, no additional assumptions.

Anatomy of the Formula

Term	Meaning
$P(A B)$	Posterior : our updated belief about A after seeing B
$P(B A)$	Likelihood : how probable is the evidence B if A were true
$P(A)$	Prior : our belief about A before seeing any evidence
$P(B)$	Evidence (marginal likelihood): total probability of observing B under all hypotheses

Intuitive reading: "Start with a belief about A (the prior). Observe evidence B. Scale that belief up or down by how well A predicts B (the likelihood). Normalise so everything sums to 1 (divide by $P(B)$). The result is your updated belief (the posterior)."

Bayes' rule is **not** a heuristic or a modelling choice — it is a **theorem** that follows directly from the axioms of probability. Any rational agent that updates beliefs on evidence must obey it.

Why Bayes Matters for Generative Models

Generative modelling: learning $p(x)$

A **generative model** is a model that learns the probability distribution $p(x)$ over some data space — images, text, audio, molecules. Once learned, we can **sample** new data by drawing from $p(x)$.

Conditional generation: learning $p(x | c)$

In practice we almost always want to generate data given a **condition** c (a text prompt, a class label, a style reference). The model must learn $p(x | c)$ — the distribution of x *given* c .

The Bayesian bridge

Apply Bayes' rule with $A = x$ (data) and $B = c$ (condition):

$$p(x | c) = p(c | x) \cdot p(x) / p(c)$$

Bayes connects $p(x | c)$ to $p(c | x)$ — enabling techniques such as classifier guidance.

The Intractability Problem

Why $P(B)$ is the bottleneck

Bayes' rule is exact, but applying it requires computing the **marginal (evidence)** $P(B)$. In continuous, high-dimensional settings this means evaluating an integral:

$$p(x) = \text{integral}_{(\text{sum over all possible latent states } z)} p(x | z) \cdot p(z) dz$$

For a latent space of even modest dimension, this integral has no closed form. The cost of numerical approximation grows **exponentially** in the dimension of z — a phenomenon known as the **curse of dimensionality**. For a 512-dimensional latent space, exact marginalisation is computationally impossible.

Coming up: autoregressive models

The autoregressive factorisation transforms an intractable joint distribution into a **sequence of tractable conditionals**. Each conditional $p(x_t | x_1, \dots, x_{t-1})$ is modelled by a neural network (e.g. a Transformer) trained with maximum-likelihood on the next token. The price paid is **sequential generation**: tokens must be sampled one at a time.